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Preliminary Result on a Dexterous Hand with Full Hand Tactile sensing and its Sim2Real Study

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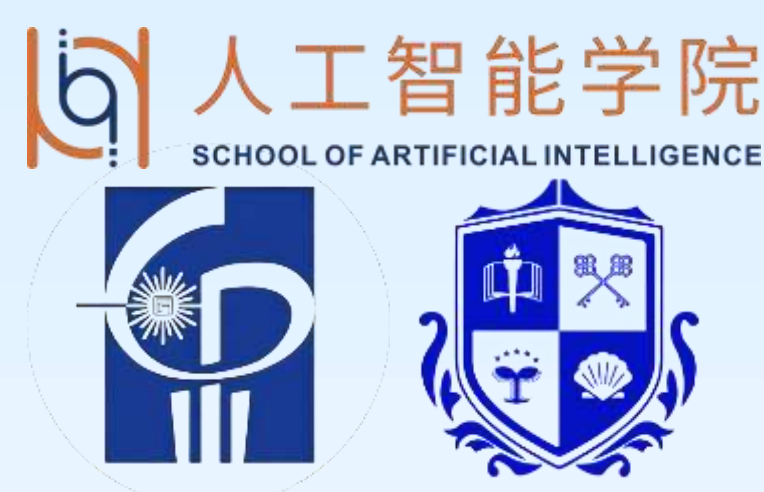
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Motivation

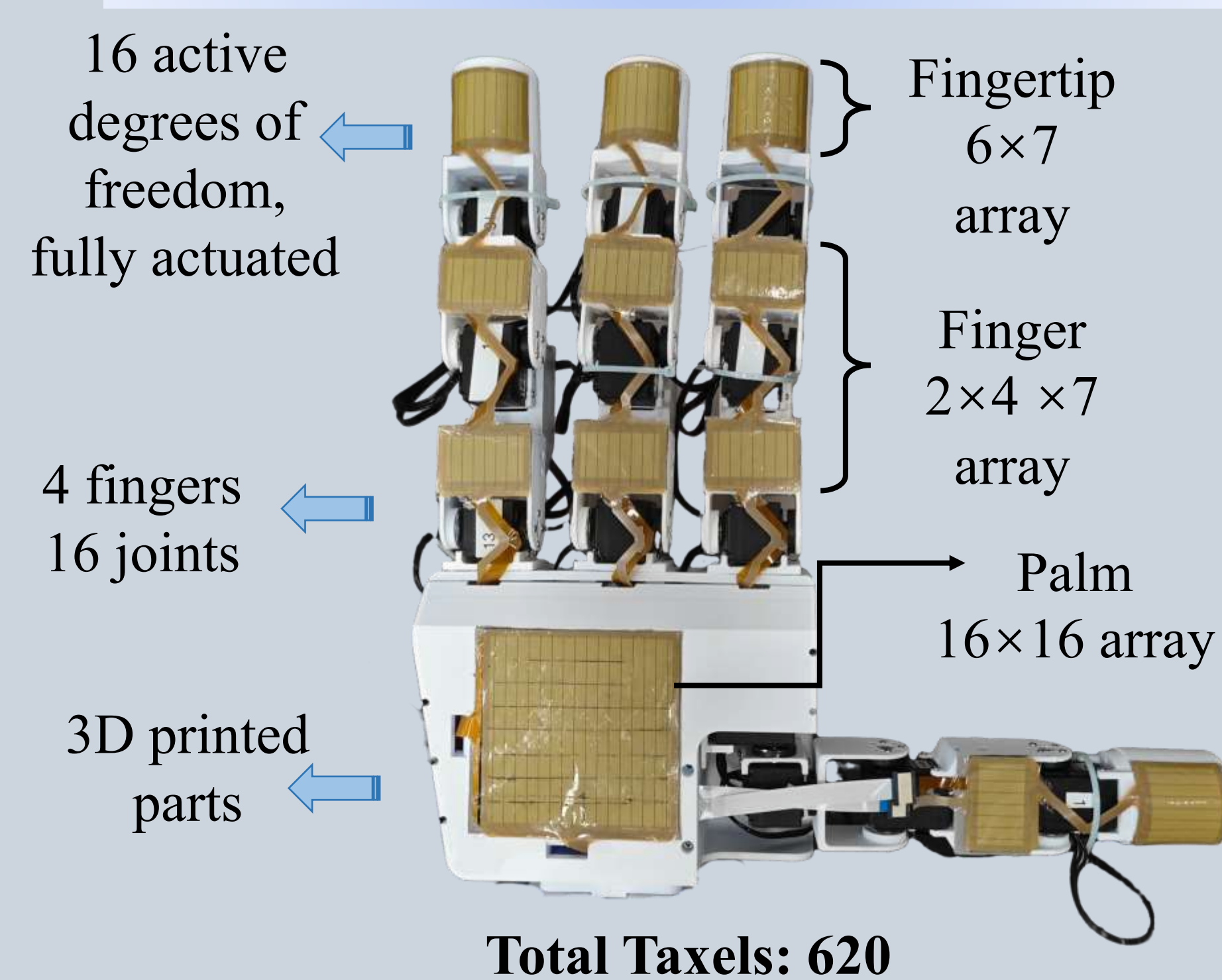


Dense full-hand tactile sensing is crucial for dexterous robotic manipulation. Existing dexterous hands lack sufficient tactile perception. Furthermore, acquiring large-scale real-world tactile data for training is challenging, making it crucial to replicate a system in simulation that aligns with the actual tactile array. To address this issue, we develop a dense tactile array dexterous hand with both real and simulation sensors, achieving a preliminary realization of tactile transfer from simulation to reality.

High-performance tactile sensor hardware: The 620-taxels fully actuated dexterous hand achieves a stable 260 Hz 750-byte full-hand tactile frame rate via 3-channel Serial Peripheral Interface (SPI) parallel acquisition and Direct Memory Access (DMA) double-buffering.

Simulation Tactile Array Align and Sim-to-Real Transfer: Raycasting combined with spring correction precisely aligns tactile taxels and normals on curved surfaces. This accurate simulation setup generates high-fidelity full-hand tactile datasets, which we use to train a learning-based three-class classification system that achieves preliminary sim-to-real transfer.

Dense tactile array dexterous hand



Methodology

FPC photolithography-fabricated piezoresistive electrode arrays (high precision, consistent response, customizable taxel resolution)

A 3-channel SPI parallel architecture cuts acquisition time by 2/3 versus single-bus, while DMA double-buffering ensures 260 Hz stability for 750-byte frames at 2 Mbps, fully parallelizing sampling and transmission.

Parameterized reading (plug-and-play heterogeneous sensors, no modification to underlying SPI acquisition logic)

Align the tactile sensor with dexterous hand model

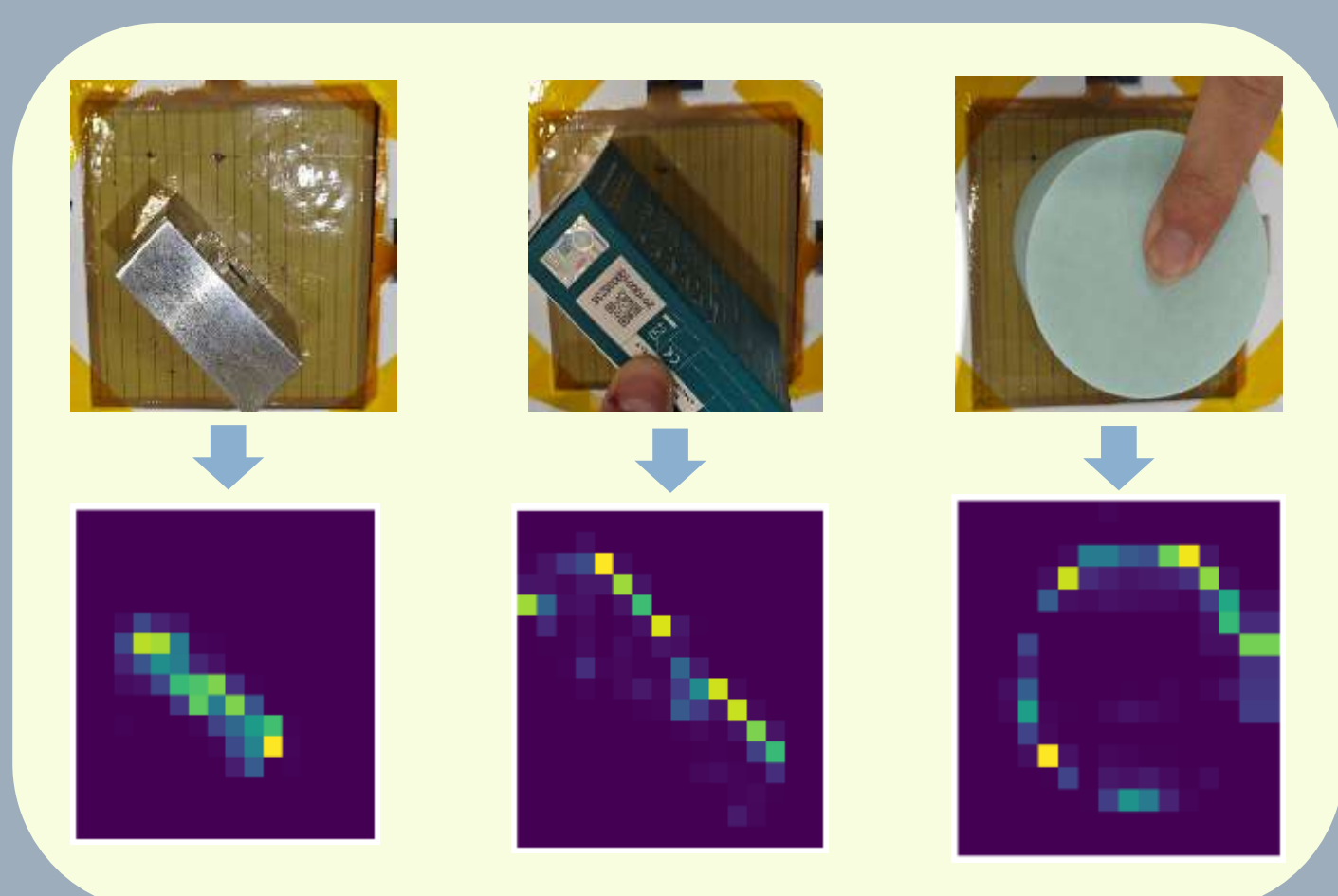
Generate planar grids on mesh tangent planes based on sensor parameters and spatial arrangement

Rays cast from grid points along mesh normals intersect triangular meshes to obtain tactile points local positions and fixed orientations

iterations of 4-neighbor spring correction

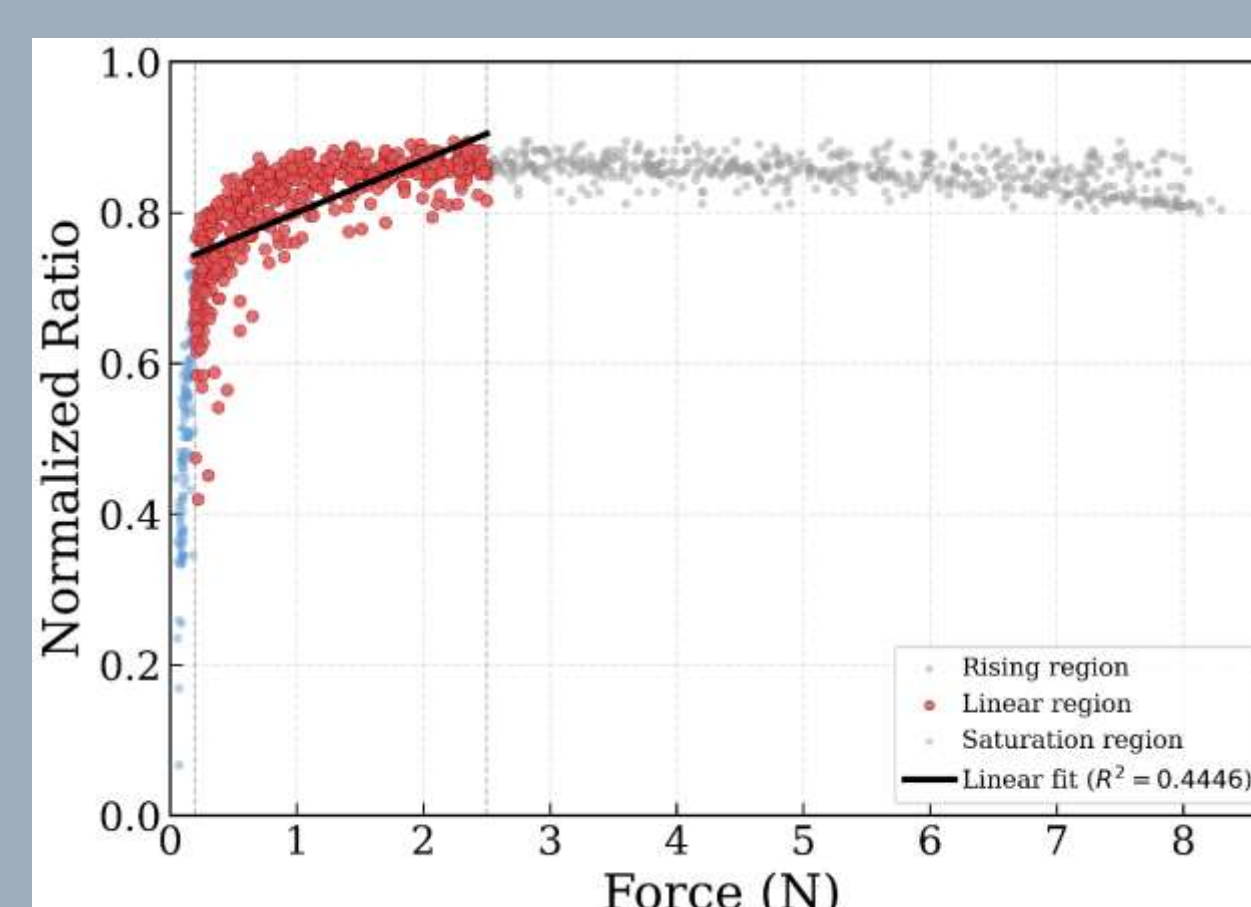
$$\mathbf{p}_i^{(k+1)} = \mathbf{p}_i^{(k)} - \lambda \sum_{j \in \mathcal{N}(i)} \frac{\mathbf{p}_i^{(k)} - \mathbf{p}_j^{(k)}}{\|\mathbf{p}_i^{(k)} - \mathbf{p}_j^{(k)}\|} \cdot (\|\mathbf{p}_i^{(k)} - \mathbf{p}_j^{(k)}\| - s)$$

Experiment

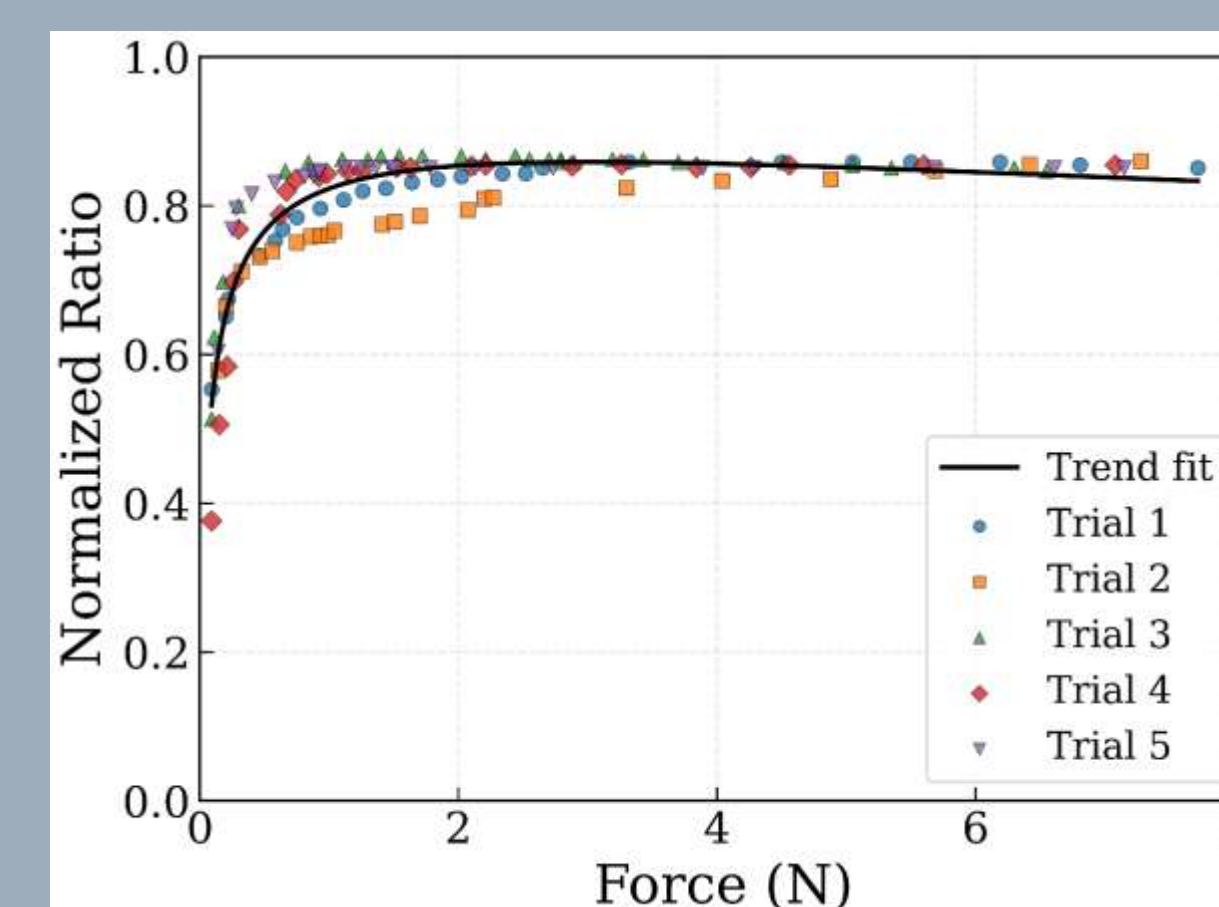


This group of subplots verifies the shape perception performance of the single tactile array sensor, presenting real-object placement scenes and synchronized contact pressure visualization results.

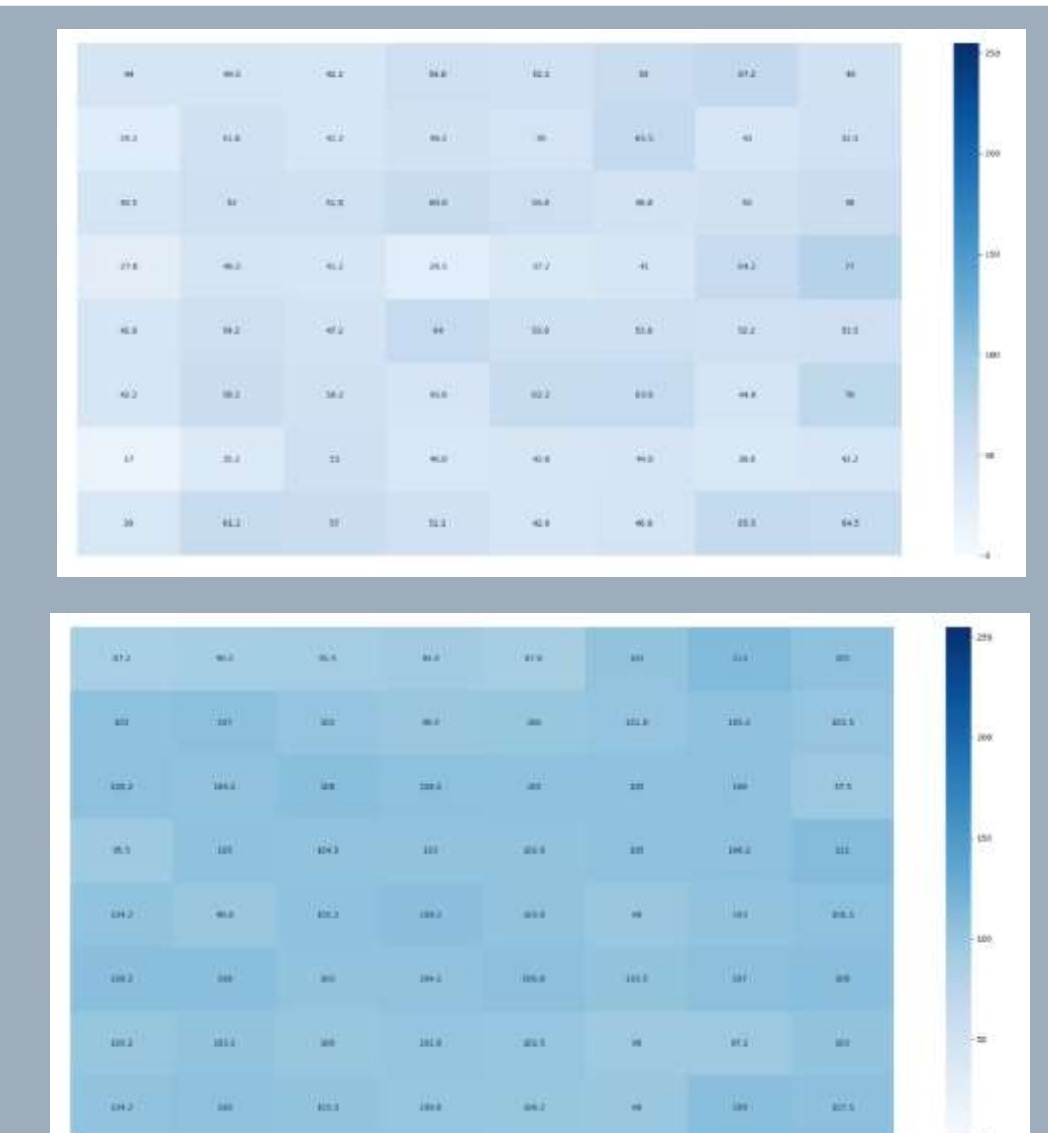
Single Tactile Sensor performance testing



(a)



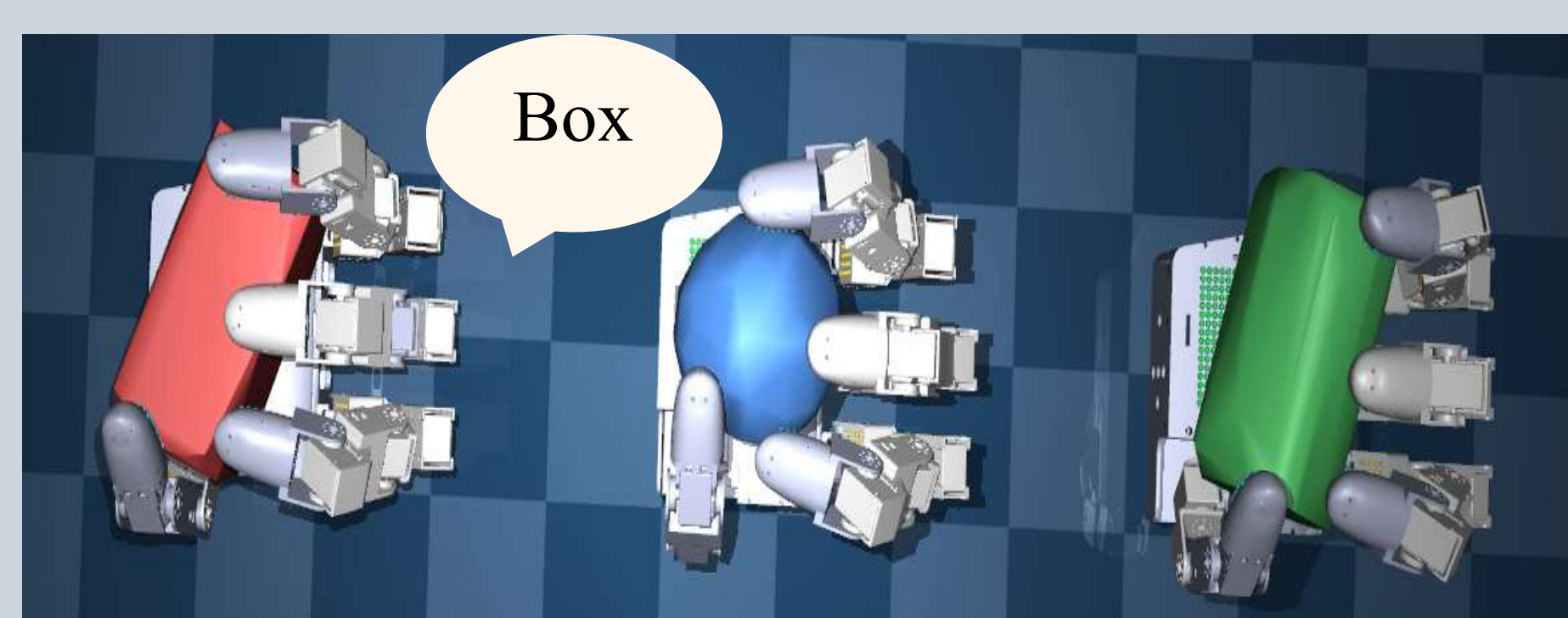
(b)



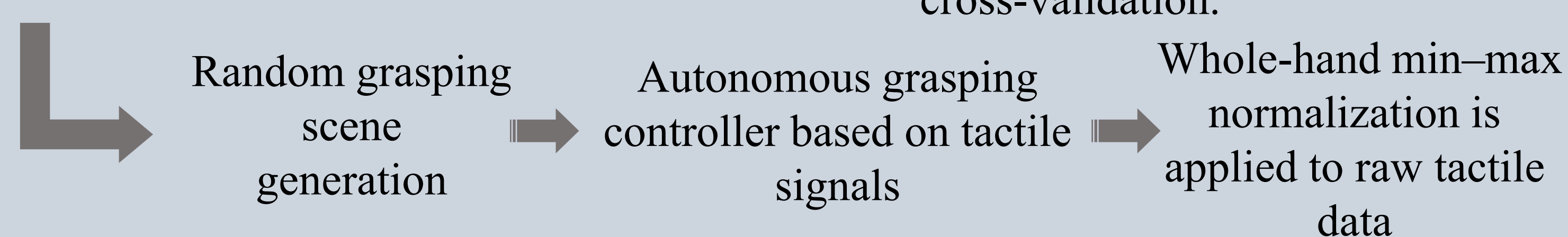
(c)

The three subplots further validate the performance of a single tactile sensor via a force gauge. Fig. a records repeated pressing data at 10 sensing points on the array, showing consistent distribution and distinct regional features, with the linear range fitted by ordinary least squares (OLS). Fig. b presents five repeated single-point tests to verify measurement consistency. Fig. c shows the two responses under simultaneous force loading on multiple array points, and the heatmap confirms uniform sensing performance.

Tactile data collection and training in Simulation



A multi-branch attention CNN is trained on a dataset of 3000 episodes collected from simulation, with label smoothing cross-entropy loss adopted for optimization. We employ Gaussian noise, random patch augmentation and 5-fold stratified cross-validation.



Model testing in Real World



	Simulation	Real World
Box	90.1	75
Sphere	90.3	70
Cylinder	89.2	80

The simulation-trained classifier deploys directly to real hardware and achieves accurate classification of 12 unseen objects without any fine-tuning, using normalized tactile signals to effectively bridge the sim-to-real gap. The table shows classification success rates on both simulated and real-world test data, validating preliminarily effective zero-shot sim-to-real transfer for tactile shape recognition. We will further extend this framework to integrate tactile recognition with dexterous manipulation capabilities in future work.

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